2014

COMPUTER SCIENCE AND ENGINEERING

Paper – CSE– 103

(Advanced Database Management Systems)

Full Marks – 100

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Answer ***Question No. 1*** and ***any four*** questions from the rest

State your assumptions clearly

1. Comment critically on the following statements and justify your opinion (***any four***) : 5×4

(a) Allocation schema is location independent although it defines the allocation of fragments for a distributed database system.

(b) The availability of a distributed system degrades gracefully in presence of increasing number of faults.

(c) Dice operation may or may not result in reduction of dimensions on an n-cube model.

(d) Client applications invoked from computers at different locations that are connected to a remote database server with a back-up server is an example of a distributed database system.

(e) Vertical fragments are typically not disjoint.

(f) Two phase locking protocol ensures consistency of data.

2. (a) Define the following :

(i) Image of a database at a site

(ii) Fragmentation schema

(iii) External schema

(iv) Mixed fragmentation

(v) Data independence.

(b) Explain the scope of vertical fragmentation towards designing of a distributed database system with a suitable example.

(c) In the context of distributed database design, why is it important to identify which part of application data are read-only items ? 10+6+4

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3. (a) Explain the significance of distributed join for query optimization in a distributed environment using an example.

(b) *Turismo* is a travel agency that books hotel rooms for its clients in big cities across India. The information to be stored is given below :

- Destination information: location id, place name, nature of place (sea-side, mountains, big cities etc.).
- Hotel information: hotel id, location id, hotel name, number of rooms, room charges.
- Guest information: guest id, name, address, phone number.
- Occupancy: hotel id, guest id, room number, check-in date, check-out date.

Design a distributed database solution for hotel room reservation system to be accessed from multiple offices of *Turismo* in multiple cities considering the following frequent queries/operations :

- Request for reservation of room. If a guest wants to book multiple rooms in same or more than one hotel for different period ; separate reservations are to be done for each stay bed and period of booking.
- Availability of room – the system is to check for availability of room in a particular hotel for a period.
- Revenue query – the system is to be used to enquire about the revenue generated in a single hotel or in a particular location or in the entire chain for a period.

Your design should include the definition of global schema, fragmentation schema and allocation schema. State your assumptions, if any. 8+12

4. (a) Describe the 6-tier architecture for a distributed database system and explain the roles for each strata.

(b) Compare the advantages and disadvantages of centralized versus distributed database systems.

(c) Explain with an example how a single updating operation on a single data item could result in multiple updating operations on several sites when the database is physically distributed.

(d) What is semi join ? How this could significantly reduce the cost of query processing in a distributed environment ? 7+4+5+4

5. (a) Explain the following OLAP operations with small examples :

(i) Roll-up

(ii) Slice

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(iii) Drill down.

(b) A cube model has data in N dimensions for a particular subject orientation. The cardinalities for the four dimensions are C_1, C_2, C_3 and C_4, respectively.

(i) How many Fact and Dimension Tables are required in an equivalent Star schema ?

(ii) State and explain the minimum number of tuples that would occur in each of Dimension and Fact Tables in your design.

(c) What is snow-flake schema ?

(d) What is concept hierarchy ? Explain with an example.

(e) Does concept hierarchy among data items on time dimension represent a completely ordered state? Explain with an example. 6+6+1+3+4

6. (a) Discuss the typical features of Semi-structured data (through an example) for which a separate data model would be needed instead of relational model.

(b) Discuss about the distinguishable properties of LOREL.

(c) How can Object Exchange Model handle the challenge of semi-structured data ?

(d) Compare ORA-SS with OEM in terms of semantic richness. 3+4+5+8

7. (a) Explain the roles of distributed transaction manager (DTM) and local transaction manager (LTM) towards transaction management.

(b) A "commit" operation in database systems ensure that the changes in the database are permanent. In the 2-phase commit protocol for transaction management, two different types of "commit" operations are performed at the coordinating site and at the participating sites. The coordinating site may decide to abort a transaction even if one or more participating site is ready for "local commit". Explain if this is consistent with the ACID properties for transaction management.

(c) "Isolation facilitates concurrent execution of transactions" — Justify the correctness of the statement.

(d) How the atomicity of a transaction is ensured under 2-phase commit protocol when one of the sites fails after the coordinator issues a global commit message ?

(e) Compare primary copy locking versus distributed 2-phase locking. 5+3+6+3+3

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